

Joseph Lavond

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EDUCATION

University of North Carolina

Ph.D. Statistics and Operations Research

M.S. Statistics and Data Science

Chapel Hill, NC

Aug. 2020 – May 2025

Aug. 2020 – May 2024

California Polytechnic State University

B.S. Statistics

San Luis Obispo

Sep. 2016 – Mar. 2020

PROFESSIONAL EXPERIENCE

Data Scientist

June 2025 – Present

Red Ventures

Fort Mill, SC

- Built and productionized ReplayOPE, an offline policy evaluation and experimentation platform for contextual bandits that surfaced stronger agent configurations in 75%+ of experiments and generated automated recommendations for production decision systems
- Evolved ReplayOPE from research proof-of-concept to internal ML product with FastAPI services, Databricks Apps, analyst-facing UI, MCP integration, usage tracking, and CI/CD via GitHub Actions and Terraform
- Developed advanced evaluation methods including combinatorially purged cross-validation, fold-based evaluation, warm starts, delayed rewards, streaming execution, concurrent runs, eligibility checks, and statistically rigorous policy comparison
- Built ReplayOPE platform infrastructure across Databricks Jobs, Terraform-managed access, Repos synchronization, automated documentation deployment, environment promotion workflows, and API route generation
- Improved core DAPI offline RL training and serving infrastructure by fixing a 3+ year missing-values inference bug, modularizing preprocessing pipelines, and expanding support for high-cardinality categorical features
- Productionized CatBoost, ordinal LightGBM, and recency-based sample weighting / historical importance strategies across DAPI agents, with several agents observing +10 percentage-point revenue lift following adoption
- Drove cross-functional adoption through analyst onboarding, product demos, searchable documentation, example workflows, and programmatic API/MCP access for LLM-driven tooling
- Rolled out a vertical-wide skills marketplace as the source of truth for approved skills and distribution, including release workflows, CI checks, LLM security checks, approval gates, and reusable bootstrapping patterns

Graduate Data Science Intern

June 2024 – Aug. 2024

Red Ventures

Fort Mill, SC

- Implemented offline model-based Q-learning for website decisioning, demonstrating a typical 5% revenue-per-session lift and estimated \$5M annual revenue opportunity
- Developed 6K+ lines of production code for a Markov Decision Process simulation environment and built a historical-production-data pipeline for larger, more complex simulation experiments

Ph.D. Intern

June 2023 – Aug. 2023

Cisco

San Jose, CA

- Developed scalable differentially private generative-model training methods and evaluated GAN mode-collapse mitigation techniques to improve private model training reliability

Graduate Info Technology Intern

June 2022 – Aug. 2022

Elevance Health

Indianapolis, IN

- Built a PyTorch semi-supervised Bayesian anomaly-detection model on AWS to identify exaggerated provider claims
- Created a PySpark SQL modeling pipeline for 60M-row claims data in Jupyter on Kubeflow; received end-of-summer presentation award

TEACHING & RESEARCH EXPERIENCE

Graduate Teaching Fellow

Spring 2022 & Fall 2024

University of North Carolina

Chapel Hill, NC

- Taught introductory statistics and data science to 100+ undergraduates across multiple semesters
- Designed student research projects on data storytelling and statistical bias to develop applied reasoning and communication skills

NSF-Funded Research Training Group

Aug. 2022 – May 2025

University of North Carolina

Chapel Hill, NC

- Selected for interdisciplinary training supported by a \$2M grant on the theory and application of networks

SELECTED PUBLICATIONS

- Joseph Lavond, Yuxin Cheng, Chudi Zhong, and Yao Li. Partition-based backdoor detection and purification from poisoned data. 2026. Under Review
- Joseph Lavond, Minhao Cheng, and Yao Li. Feddecay: Balancing model performance and rapid personalization in federated learning with learning rate scheduling. *Journal of Computational and Graph Statistics*, 2026. Accepted
- Joseph Lavond. *Advancing Model Security, Data Privacy, and Performance for Widespread Adoption of Trustworthy Artificial Intelligence*. PhD thesis, University of North Carolina at Chapel Hill, 2025
- Joseph Lavond, Minhao Cheng, and Yao Li. Trusted aggregation (TAG): Backdoor defense in federated learning. *Transactions on Machine Learning Research*, 2024

SELECTED PROJECTS

Statistical Consultant, University of North Carolina

Aug. 2022 – Dec. 2023

- Used R to analyze longitudinal behavioral and operational data, identifying early autism-risk markers and key drivers of medical contract turnaround delays

Fatal Force, University of North Carolina

Aug. 2020 – May 2021

- Engineered features from FBI crime reports and Census data and evaluated statistical models predicting police shooting fatalities, improving test RMSE by 3%

Frost Summer Research Program, Cal Poly

May 2019 – Sep. 2019

- Built an R Shiny survival-analysis app for cancer prognosis using KM and Cox models and processed SEER data covering 20M cancer cases

SKILLS

ML & Decision Systems: Contextual Bandits, Offline Policy Evaluation (OPE), Reinforcement Learning, Policy Evaluation, LightGBM, CatBoost, Ordinal Encoding, Sample Weighting, Historical Importance, Ensemble Methods, scikit-learn

Statistics & Deep Learning: Bayesian Inference, Experiment Design, Cross-Validation, Purged Cross-Validation, A/B Testing, Statistical Testing, Policy Comparison, PyTorch, TensorFlow, Keras

ML Systems & AI Infrastructure: FastAPI, Databricks Apps/Jobs, DBRX, GitHub Actions, Terraform, CI/CD, Model Deployment, API Design, MCP, LLM Workflows, Skills Marketplaces

Programming & Data: Python, SQL, R, Pandas, NumPy, PySpark, SciPy, Tidyverse, Feature Engineering, Data Pipelines, Data Quality & Validation

Analytics, Visualization & Tools: Clustering, Segmentation, Dashboarding, Matplotlib, Seaborn, ggplot2, Git, Linux